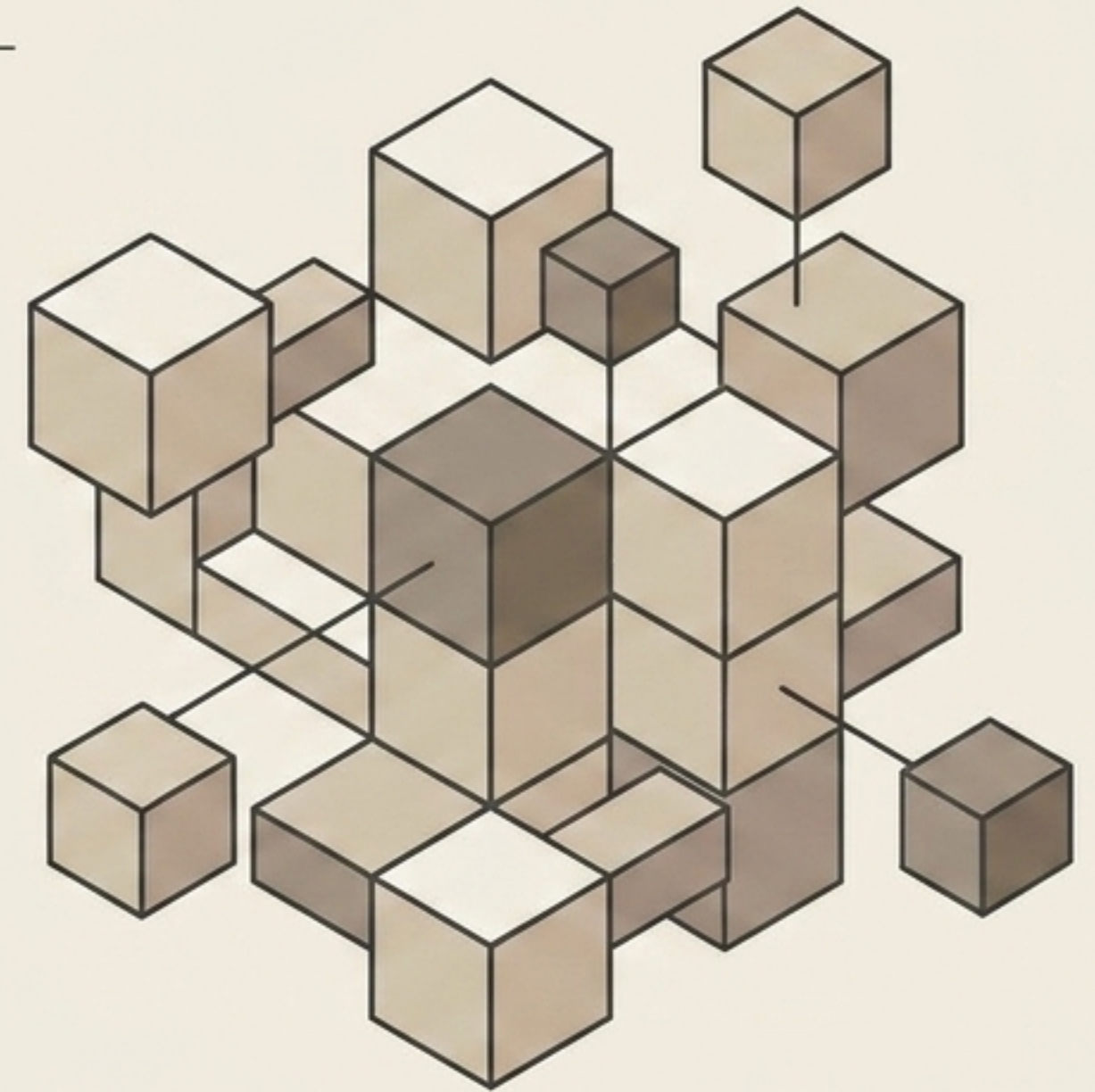


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AI Product Discovery Platform

Database Design

Batch 4 - Group 8

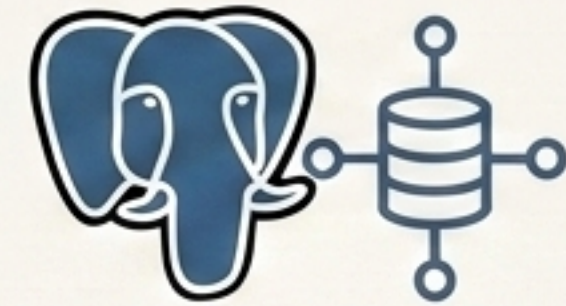


Database Overview



Supabase Engine

Single source of truth for the entire idea-validation workflow.



PostgreSQL Core

Strict relational integrity combined with complex JSONB data support.



System Role

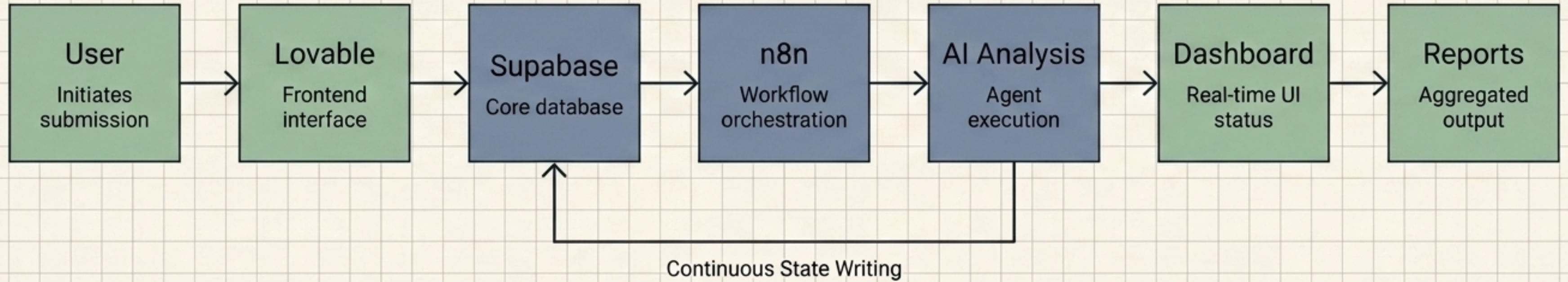
Stores user profiles, raw submissions, asynchronous AI analyses, and confidence scores.



Workflow Pattern

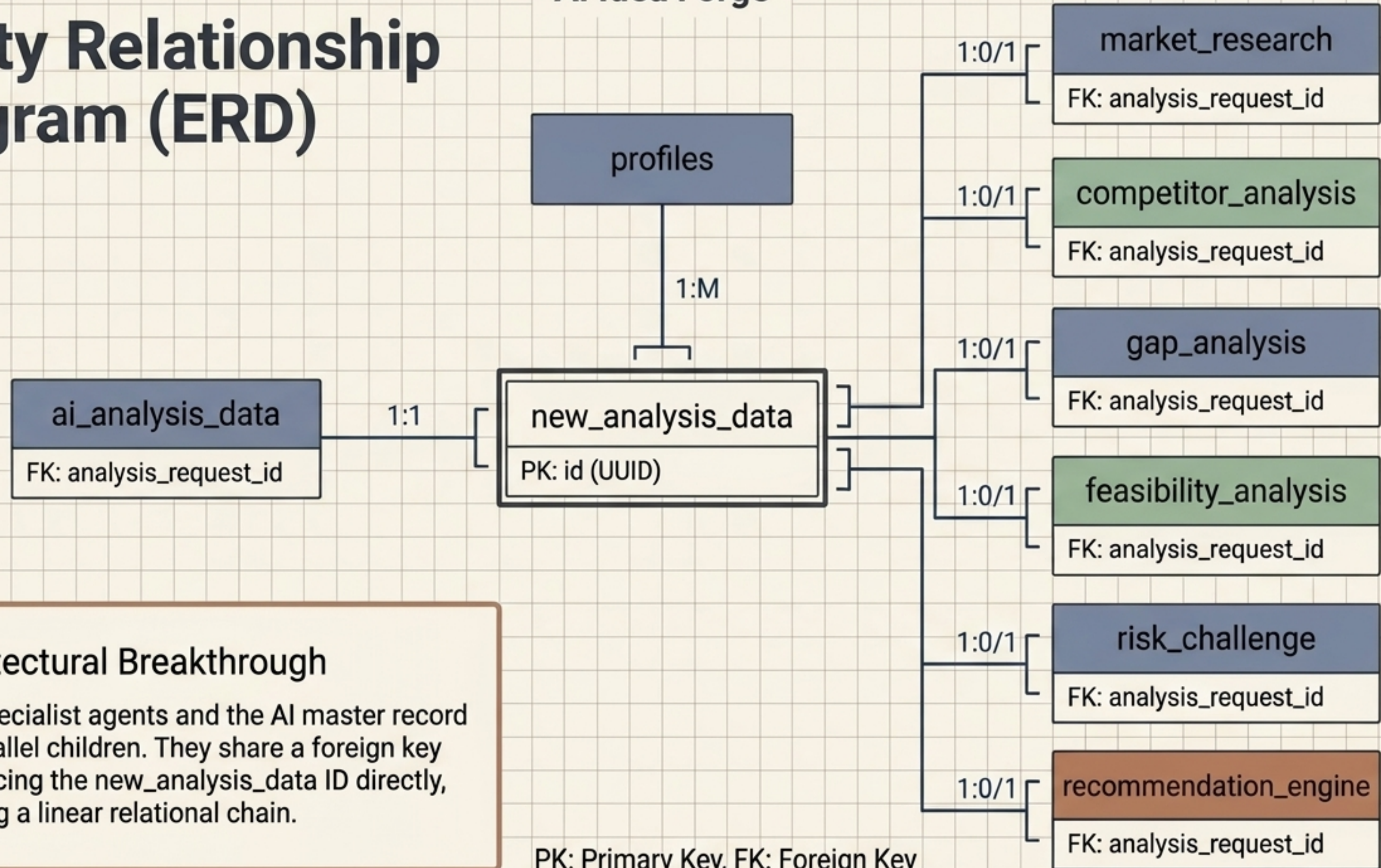
Idea Submission to AI Master Record to 6 Specialist Agents to Aggregated Reports.

Database Architecture



Entity Relationship Diagram (ERD)

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Architectural Breakthrough

All 6 specialist agents and the AI master record are parallel children. They share a foreign key referencing the **new_analysis_data** ID directly, avoiding a linear relational chain.

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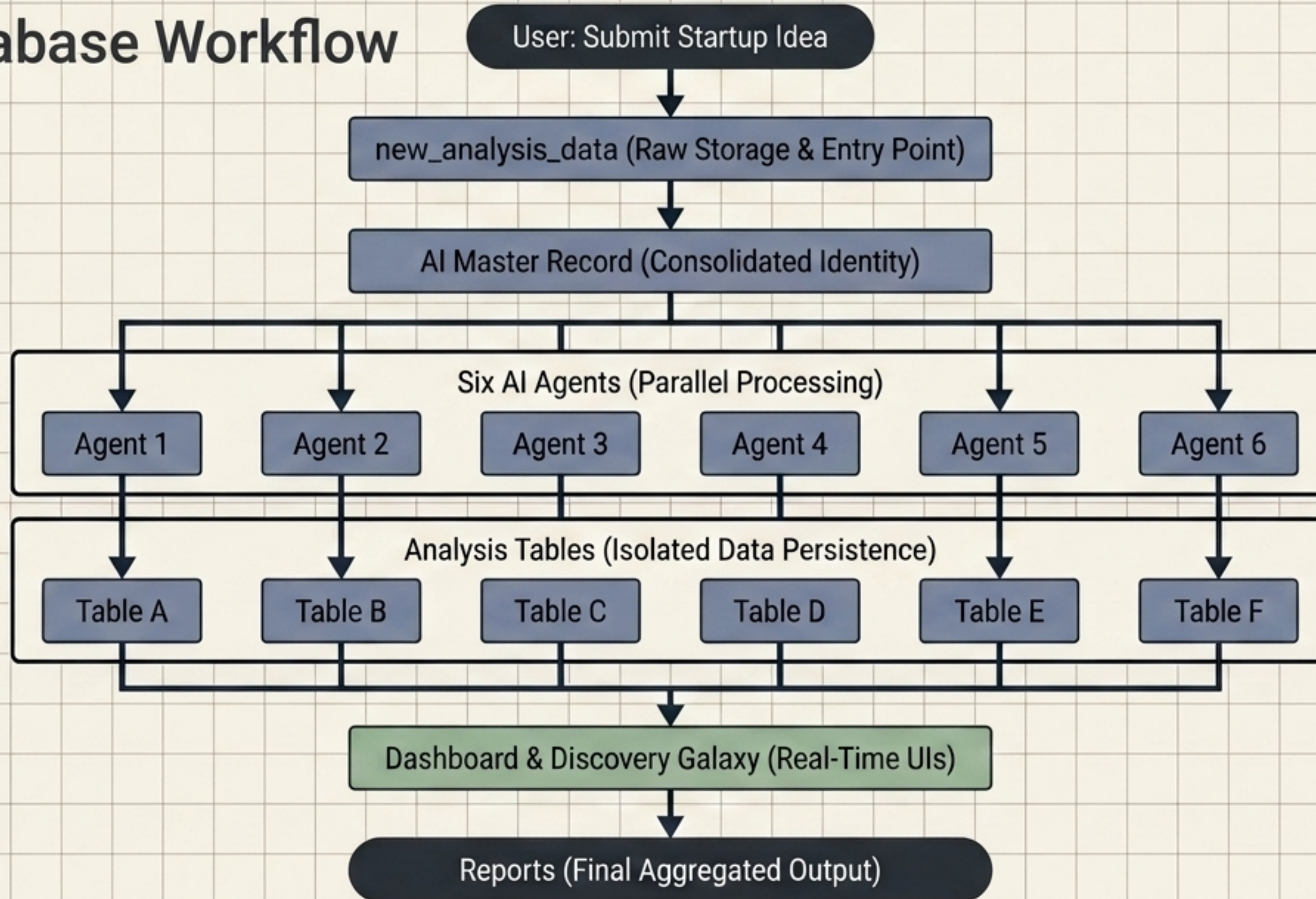
Database Relationship Summary

Parent Table	Child Entity	Relationship	Architectural Rationale
profiles (User)	new_analysis_data	One-to-Many	Users test multiple startup ideas over time.
new_analysis_data	ai_analysis_data	One-to-One	Creates one consolidated identity record per idea.
new_analysis_data	[Six Specialist AI Agent Tables]	One-to-Zero/One	Asynchronous generation; child rows do not exist until the respective agent successfully completes analysis.

Parallel Architecture ensures all AI outputs reference the parent submission directly via analysis_request_id.

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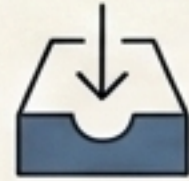
Core Database Workflow



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Core Tables

new_analysis_data



Purpose: Entry point; immutably stores the raw, unprocessed idea submission.

Primary Key: id (UUID)

Foreign Key: user_id (References user profile)

Used By: Idea Submission Form, Dashboard UI

ai_analysis_data



Purpose: AI-generated master identity (refined title, industry classification, matching score).

Primary Key: id (UUID)

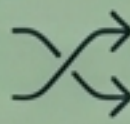
Foreign Key: analysis_request_id (References raw submission)

Used By: Discovery Galaxy (Header), Resume Analysis logic

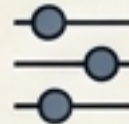
AI Agent Tables

All tables share the exact same PK/FK structure: *id* (UUID) and *analysis_request_id* (FK).

	Market Research
Output: Market size, trends, barriers.	
Used By: Galaxy / Reports	

	Competitor Analysis
Output: Direct/indirect competitors, positioning.	
Used By: Galaxy / Reports	

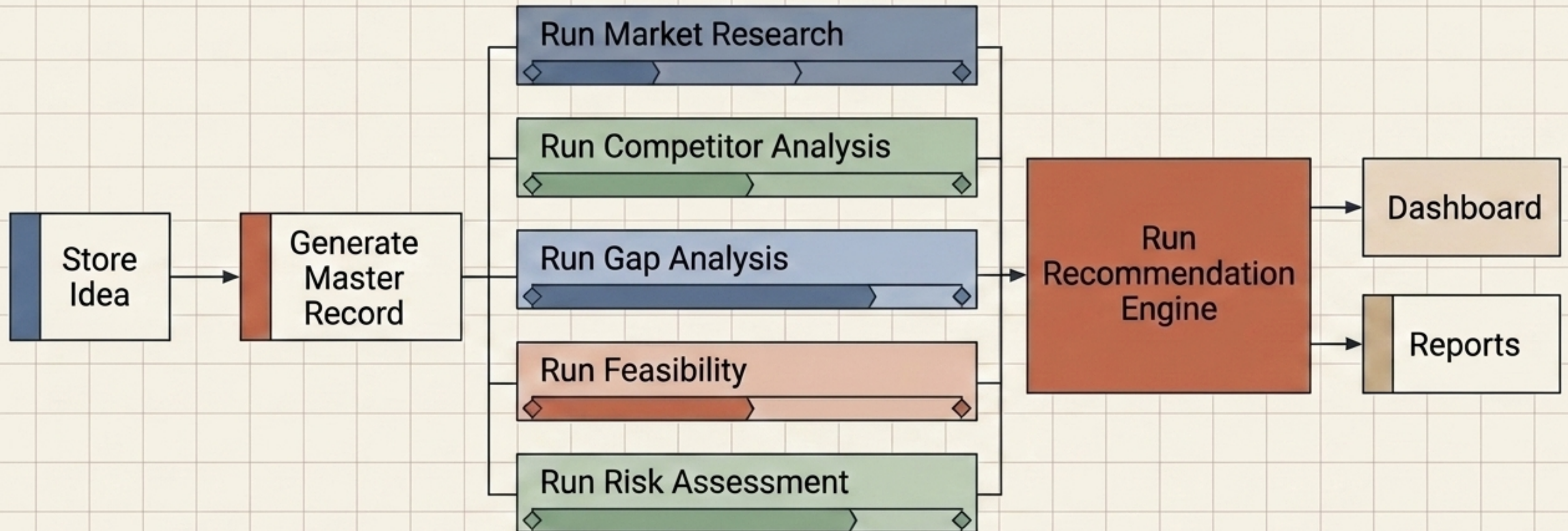
	Gap Analysis
Output: Unmet needs, innovation opportunities.	
Used By: Galaxy / Reports	

Feasibility Analysis	
Output: Tech/Business scores, resources.	
Used By: Galaxy / Reports	

Risk & Challenge	
Output: Market/Financial risks, mitigations.	
Used By: Galaxy / Reports	

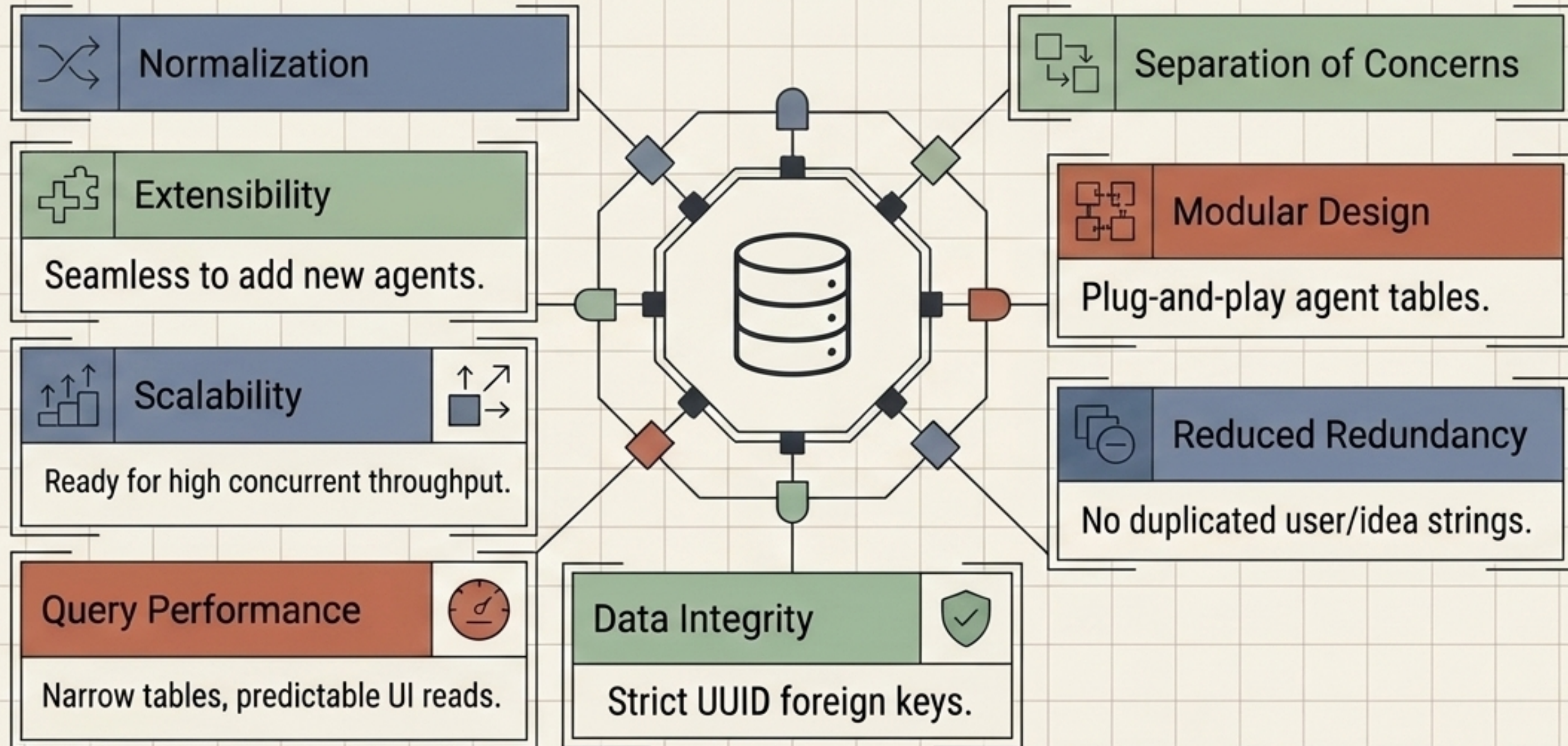
Recommendation Engine	
Output: Opportunity score, strategic next steps.	
Used By: Galaxy, Dashboard	

Database Workflow (Execution)

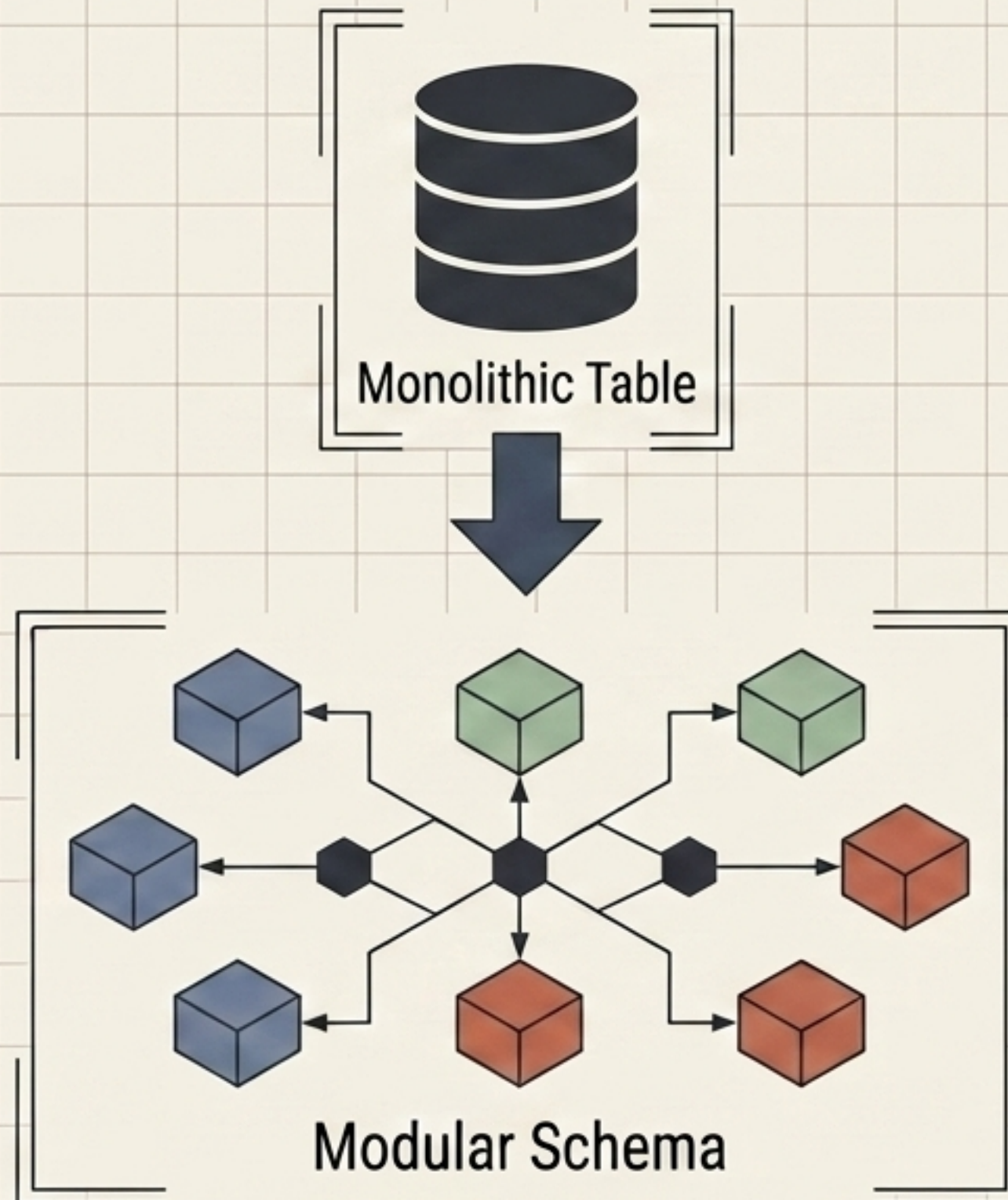


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Database Design Principles



Benefits of the Database Design



AI Agent Isolation

Schema for one agent can evolve without breaking others.

Flexible & Scalable

Handles asynchronous AI generation flawlessly without lockouts.

Easier Future Expansion

Adding a new agent requires only one new child table.

Better Performance

UIs query narrow, specific tables instead of filtering one overloaded mega-table.

Consistent Data Management

Predictable, standardized Foreign Key patterns across the entire application.

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Architectural Summary:

- Database schema perfectly supports the asynchronous AI workflow.
- Modular isolation simplifies enterprise maintenance.
- Agent separation ensures maximum horizontal scalability.
- A solid, scalable foundation established for future AI capabilities.

Thank You

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